



Technical Service Bulletin

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New Aftercooler Assemblies

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Technical Service Bulletin announces new aftercooler manifolds and aftercooler covers for use on QSK60 diesel engines. The new aftercooler assemblies are manufactured from cast iron to increase the durability of the components.

The new cast iron aftercooler assemblies can be fitted to all QSK60 diesel engines which presently contain the aluminium manifolds, however they are **not** compatible with Phase 1.0 or Phase 1.5 CENSE™ harnesses or vented aftercooler cores. If engines with Phase 1.0 or Phase 1.5 CENSE™ harnesses are upgraded with the cast iron manifolds, the entire engine wiring harness will require an upgrade to a Phase 2 wiring harness. If engines with vented cores are upgraded, the core will also require changing. Table 5 list all the required parts for a CENSE™ engine wiring harness upgrade.

Note : Engines without CENSE™ do not require engine harness upgrades, however a standard engine wiring harness and sensor parts list can be found in Table 4, if required.

Cast iron aftercooler components **must** be changed as a quadrant assembly. Cast iron covers **must only** be assembled to cast iron manifolds and aluminium covers **must only** be assembled to aluminium manifolds. This is due to the difference in thermal expansion rates of the two materials. Mixing cast iron and aluminium covers and manifolds can result in air leaks. Cast iron and aluminium aftercooler quadrant assemblies may be fitted to the same engine.

This parts announcement does **not** obsolete the present aluminium aftercooler assemblies. These will still be available as they are required for use on QSK60 Fracture Rig engines.

Cast iron aftercoolers went in to production on QSK60 industrial engines, except for Fracture Rig engines, at Daventry Engine Plant, as of 03 March 2008, with an engine serial number (ESN) first of 33172728. QSK60 Fracture Rig and Power Generator engines are still manufactured using aluminium aftercooler assemblies.

The new cast iron aftercooler assemblies are heavier than the aluminium aftercooler assemblies and therefore require a higher tensile steel capscrews with greater torque capability. All the required new capscrews are listed in Table 2. Use of the aluminium aftercooler assembly mounting capscrews on the cast iron aftercooler assemblies can lead to capscrew breakage and possible engine damage or performance issues.

Note : The torque values for the cast iron aftercooler assembly mounting hardware are greater than that of the aluminium aftercoolers. Refer to Procedure 010-002 (Aftercooler Assembly) in the QSK45 and 60 Service Manual, Bulletin 4021530, for further information on required torques and aftercooler assembly instructions. (</qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html>) Follow the instructions in this procedure; the increased weight of the cast iron aftercooler assemblies can cause personal injury.

Table 1 contains the new cast iron aftercooler part numbers and the location on the engine. Table 2 contains the related parts which **must** be installed at the same time as the cast iron aftercooler manifolds and covers.

Table 1: Cast Iron Aftercooler Manifolds and Covers

Name	Part Number	Description/Location	Drawing Number
Manifold, Air	4999047	Right Bank Rear/Left Bank Front - Manifold	39
Cover, Aftercooler	4999054	Right Bank Rear/Left Bank Front - Cover	37
Manifold, Air	4999050	Right Bank Front/Left Bank Rear - Manifold	40
Cover, Aftercooler	4999052	Right Bank Front/Left Bank Rear - Cover	38

Table 2: New Related Parts (per quadrant)

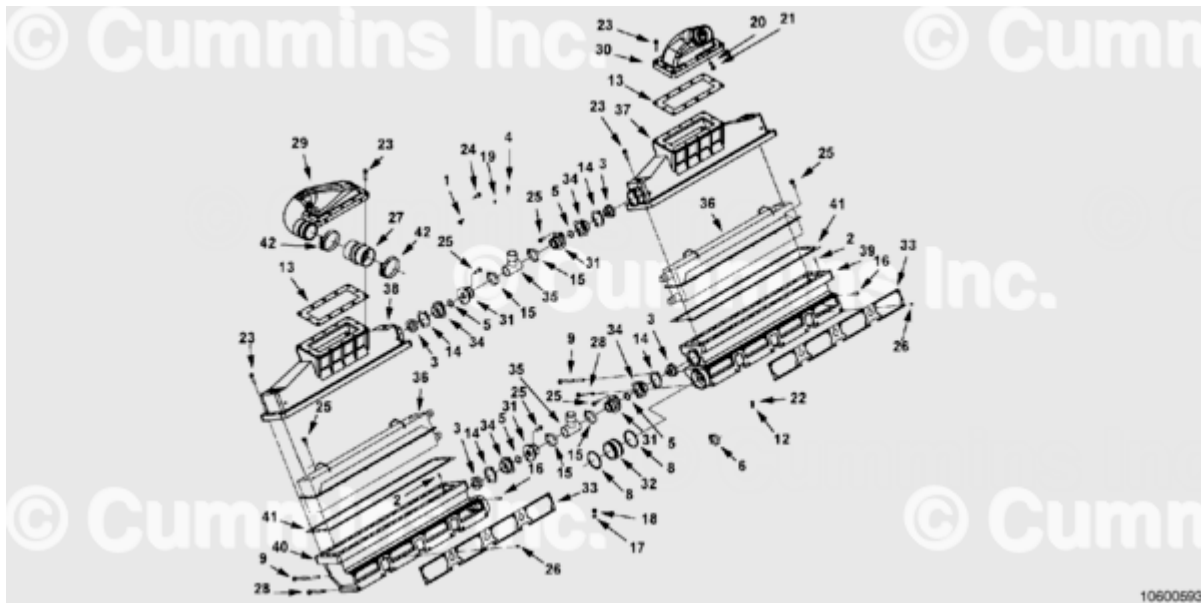
Name	Part Number	Description/Location	Quantity (per quadrant)	Drawing Number
Screw, Hex Flange Head Cap	3681390	Manifold Capscrew	8	28
Screw, Hex Flange Head Cap	4924790	Manifold Capscrew	8	9
Screw, Hex Flange Head Cap	3093932	Aftercooler Core Capscrew	12	25
Screw, Hex Flange Head Cap	3925437	Aftercooler Cover Capscrew	37	23

Table 2: New Related Parts (per quadrant)				
Name	Part Number	Description/Location	Quantity (per quadrant)	Drawing Number
Screw, Hex Flange Head Cap	3102651	Aftercooler Cover Capscrew	2	23
Gasket, Intake	3637776	Manifold Gasket	1	33
Plug, Threaded	3678921	Plug	2	

All other aftercooler assembly components are the same on cast iron as on the aluminium aftercooler assemblies. The following table contains the common aftercooler components:

Table 3: Common Parts (per bank, left or right bank)			
Name	Part Number	Quantity (per quadrant)	Drawing Number
Pin, Roll	131584	8	2
Seal, Grommet	206808	4	3
Seal, O	3035026	4	5
Gasket, Connection	3171170	2	13
Gasket, Connection	3171287	4	14
Seal, Rectangular	3331834	16	16
Plug, Pipe	3628604	2	19
Plug, Pipe	3628605	2	
Hose, Hump	3635366	2	27
Clamp, Hose	3637289	4	15
Screw, Hex	3900634	24	23
Screw, Hex	3903112	8	25
Screw, Hex	3921851	4	26
Flange, Connection	4017163	4	31
Retainer, Seal	4066129	4	34
Gasket, Aftercooler	4080573	2	41
Core, Aftercooler	4101030	2	36
Clamp, Hose	3043487	1	6
Seal, O	3082358	1	8
Clamp, Hose	3254315	4	15
Draincock	3638582	1	1
Nipple, Coupling	3638769	2	
Connection, Water	4066279	1	35

Note : The common part quantities listed above are for a single bank rebuild. All the components listed in Table 1, Table 2 (x2) and Table 3 are required for a complete left bank or right bank rebuild as illustrated in the graphic. For a complete engine rebuild double all the quantities listed above.



The following tables contain the required parts for upgrading to a Phase 2 engine wiring harness.

Note : This is only required when engines with Phase 1.0 or Phase 1.5 CENSE™ harnesses upgrade from aluminum to cast iron aftercooler assemblies.

Table 4: Phase 2 Standard Engine Harness and Sensors		
Name	Part Number	Quantity
Kit, Engine Harness	4090061	1
Timing Rail Pressure (top)	3408564	1
Fuel Temperature (middle)	3865346	1
Fuel Rail Pressure (bottom)	3408562	1
Intake Manifold Temperature	3408345	1
Intake Manifold Pressure	3408556	1
Fuel Pump Pressure	3408566	1
Coolant Temp	3865346	1
Coolant Pressure	3408556	1
Oil Level	3408653	1
Oil Rifle Pressure (gauge)	4088734	1
Speed Sensors '0' FWH	3408472	1
Speed Sensors '00' FWH	3408471	1
Blowby Pressure	3867561	1

Table 4: Phase 2 Standard Engine Harness and Sensors		
Name	Part Number	Quantity
Amb Air Pressure	3085140	1
Table 5: Phase 2 CENSE™ Engine Harness and Sensors		
Name	Part Number	Quantity
Kit, Engine Harness	4090060	1
Timing Rail Pressure (top)	3408564	1
Fuel Temperature (middle)	3865346	1
Fuel Rail Pressure (bottom)	3408562	1
Intake Manifold Temperature	3408345	1
Intake Manifold Pressure	3408556	1
Fuel Pump Pressure	3408566	1
Coolant Temp	3865346	1
Coolant Pressure	3408556	1
Oil Level	3408653	1
Oil Rifle Pressure (gauge)	4088734	1
Speed Sensors '0' FWH	3408472	1
Speed Sensors '00' FWH	3408471	1
Blowby Pressure	3867561	1
Amb Air Pressure	3085140	1
Oil Temp	3865346	1
Turbo-Compressor Inlet Temperature	3408345	1
Lub Oil Pressure (post filter)	3408558	1
Lub Oil Pressure (pre filter)	3408558	1
EGTS (shorter)	3408543	14
EGTS (longer)	3408544	2